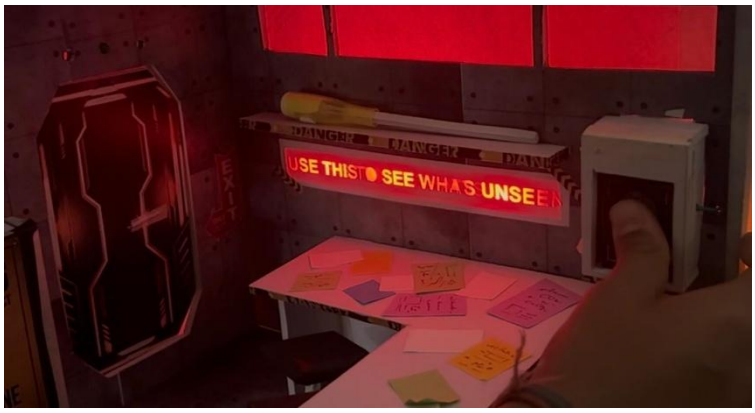
1. Test Plan

Test	Method	Expected Result
Room 1 Puzzle Logic	User decodes the hint and presses the button	The correct button activates the green LED and completes Room 1
Room 2 Sensor Logic	User triggers the wire sensor and the IR sensor	The servo opens next door successfully
Room 3 Final Challenge	User presses the correct final button	Buzzer stops, red LED turns off, green LED turns on
LED Response Time	Observe the output after the input is given	LEDs respond instantly or within 1 second
Servo Performance	Repeated door opening tests	Servo rotates smoothly without delay
Buzzer Functionality	Trigger the final room alarm	Clear sound output during the challenge
Wiring Stability	Move the model slightly during testing	Circuit continues functioning without loose connections
Overall Gameplay Flow	The full game runs from Room 1 to Room 3	Smooth progression through all levels



2. User Testing

Who were the users?

i. *Classmates*
ii. *Friends*
iii. *Team members during internal testing*
iv. *Faculty during final demo practice*

What tasks were given?

i. *Understand the storyline and mission*
ii. *Solve Room 1 puzzle using clues*
iii. *Progress through Room 2 using sensors*
iv. *Complete final room under pressure*
v. *Give feedback on difficulty and experience*

3. Results

Test	Outcome	Pass/Fail
Room 1 Puzzle	*Puzzle logic worked correctly after following the hints, though wire connectors failed once or twice during repeated testing*	*Partial Pass*
Room 2 Door Mechanism	*Servo successfully opened the door, but occasional interruptions and inconsistent movement occurred during some trials*	*Partial Pass*
Room 3 Final Alarm	*The correct button stopped the buzzer and activated the green LED as intended*	*Pass*
LED Response	*LEDs responded quickly and clearly after correct inputs*	*Pass*
Wiring Stability	*Loose wire connections caused temporary circuit failures during testing*	*Partial Pass*
Full Gameplay Run	*The entire game sequence worked successfully after debugging and reconnecting components*	*Pass*

4. Observations

User Behavior

- *Users were curious and engaged by the three-room progression.*
- *Most users followed LED clues naturally.*
- *The final room created excitement due to the buzzer and urgency.*

Pain Points

- *Some users needed hints in Room 1 initially.*
- *Loose wires occasionally interrupted gameplay.*
- *IR sensor placement needs correct positioning for detection.*
- *Some puzzles were solved too quickly once clues were understood.*

6. Improvements Suggested

- *Add an in-built timer within the code itself to create stronger urgency and automate time tracking during gameplay.*
- *Provide a subtle clue or indicator in the final room to help players identify the correct button while still maintaining challenge.*
- *Reconstruct and secure wiring connections to avoid loose circuits and improve overall reliability.*
- *Improve internal cable management for a cleaner and more organised setup.*
- *Plan puzzle flow earlier in the process so testing and integration can happen more smoothly.*
- *Increase time for trial runs before final presentation to identify issues in advance.*